

Juan Carlos Verduzco

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West Lafayette, IN

EDUCATION

Doctor of Philosophy – Materials Engineering

Purdue University, West Lafayette, IN

2018 – 2023

Bachelor of Science – Mechatronics Engineering

Monterrey Institute of Technology and Higher Education (ITESM), Zapopan, MEX

Study Abroad program on Artificial Intelligence at Universitat Politècnica de Catalunya (Spain)

2013 – 2017

PROFESSIONAL EXPERIENCE

Lead Software Engineer

Purdue University, West Lafayette, IN

2026 – Present

Senior Data Science Engineer

Purdue University, West Lafayette, IN

2023 – 2026

Graduate Research Assistant

Purdue University, West Lafayette, IN

2018 – 2023

Test Engineering Technician III

Jabil Circuit Mexico, Zapopan, MEX

2018 – 2018

PUBLICATIONS

Peer-Reviewed Articles

- [1] Mohnish Harwani, Juan Carlos Verduzco, Brian H Lee, and Alejandro Strachan. "Accelerating active learning materials discovery with FAIR data and workflows: a case study for alloy melting temperatures". In: *Computational Materials Science* 249 (2025), p. 113640.
- [2] Chunyu Li, Juan Carlos Verduzco, Brian H Lee, Robert J Appleton, and Alejandro Strachan. "Mapping microstructure to shock-induced temperature fields using deep learning". In: *npj Computational Materials* 9.1 (2023), p. 178.
- [3] Juan Carlos Verduzco, Brian Bettes, Qixuan Hu, and Ernesto E Marinero. "A review on defect engineering of anode materials for solid-state battery applications". In: *Ionics* 29.2 (2023), pp. 439–453.
- [4] Juan Carlos Verduzco, Ernesto E Marinero, and Alejandro Strachan. "Atomistic mechanisms underlying the maximum in diffusivity in doped $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ". In: *The Journal of Physical Chemistry C* 127.22 (2023), pp. 10806–10812.
- [5] David E Farache, Juan Carlos Verduzco, Zachary D McClure, Saaketh Desai, and Alejandro Strachan. "Active learning and molecular dynamics simulations to find high melting temperature alloys". In: *Computational Materials Science* 209 (2022), p. 111386.
- [6] Prateek Malhotra, Juan Carlos Verduzco, Subhayan Biswas, and Ganesh D Sharma. "Active discovery of donor: acceptor combinations for efficient organic solar cells". In: *ACS Applied Materials & Interfaces* 14.49 (2022), pp. 54895–54906.
- [7] Juan Carlos Verduzco, Ernesto E Marinero, and Alejandro Strachan. "An active learning approach for the design of doped LLZO ceramic garnets for battery applications". In: *Integrating Materials and Manufacturing Innovation* 10.2 (2021), pp. 299–310.
- [8] Juan Carlos Verduzco, John N Vergados, Alejandro Strachan, and Ernesto E Marinero. "Hybrid polymer-garnet materials for all-solid-state energy storage devices". In: *ACS omega* 6.24 (2021), pp. 15551–15558.
- [9] Andres Villa, Juan Carlos Verduzco, Joseph A Libera, and Ernesto E Marinero. "Ionic conductivity optimization of composite polymer electrolytes through filler particle chemical modification". In: *Ionics* 27 (2021), pp. 2483–2493.

Preprints

- [1] Thomas M Deucher, Juan Carlos Verduzco, Michael Titus, and Alejandro Strachan. "A collaborative digital twin built on FAIR data and compute infrastructure". In: *arXiv preprint arXiv:2507.00048* (2025).
- [2] Juan Carlos Verduzco, Mohnish Harwani, Daniel Mejia, and Alejandro Strachan. "Autonomous Universal Research Assistant (AURA): Agentic AI meets nanoHUB's FAIR Workflows and Data". In: *chemRxiv preprint chemRxiv: 10.26434/chemrxiv-2025-rs07k* (2025).
- [3] Daniel Mejia, Steven Clark, Juan Carlos Verduzco, Michael Zentner, Lynn Zentner, Gerhard Klimeck, and Alejandro Strachan. "nanoHUB services for FAIR simulations and data: ResultsDB and Sim2Ls". In: *arXiv preprint arXiv:2306.11062* (2023).
- [4] Juan Carlos Verduzco, Ethan Holbrook, and Alejandro Strachan. "GPT-4 as an interface between researchers and computational software: improving usability and reproducibility". In: *arXiv preprint arXiv:2310.11458* (2023).

Conference Presentations and Proceedings

- [1] Ethan Holbrook, Juan Carlos Verduzco, Kat Nykiel, William Zummo, and Alejandro Strachan. "AI-Powered Interface: Fully Automated Tool for LAMMPS Simulation and Analysis". In: *TMS2025 Technical Program*. 2025.
- [2] Ethan Holbrook, Juan Carlos Verduzco, and Alejandro Strachan. "Natural Language to LAMMPS: LLMs as interfaces between researchers and scientific software". In: *LAMMPS Workshop and Symposium*. 2025.
- [3] Juan Carlos Verduzco, Ethan Holbrook, and Alejandro Strachan. "Teaching AI to Speak LAMMPS: A Practical Guide to Prompting and Script Checking". In: *LAMMPS Workshop and Symposium*. 2025.
- [4] Juan Carlos Verduzco and Alejandro Strachan. "nanoRA: A Multi-Agent Research Assistant for Autonomous Scientific Workflows in nanoHUB". In: *Artificial Intelligence and Machine Learning for Materials: State of the Art and Outlook (Workshop)*. 2025.
- [5] Ernesto Marinero, Sebastian Calderon, Juan Carlos Verduzco, Javier Castillo, Regina Park, Praveen Balaji, Yaisa Ramirez, and Alex Wei. "Li-salt dissociation and polymer microstructural changes in composite polymer electrolytes". In: *APS March Meeting Abstracts*. Vol. 2024. 2024, G64–003.
- [6] Juan Carlos Verduzco, Sebastian Calderon Cazorla, Gavin Bidna, Alexander Wei, Alejandro Strachan, and Ernesto Marinero. "High entropy oxides: promising fillers for high ionic conductivity composite polymer electrolytes". In: *APS March Meeting Abstracts*. Vol. 2023. 2023, G45–005.
- [7] Juan Carlos Verduzco, Daniel Mejia, Steven Clark, David Farache, and Alejandro Strachan. "Towards FAIR Simulation Workflows: nanoHUB's Sim2Ls and ResultsDB". In: *TMS2023 Technical Program*. 2023.
- [8] Juan Carlos Verduzco and Alejandro Strachan. "Exploring the Compositional Space of High Entropy Alloys via Sequential Learning". In: *TMS2023 Technical Program*. 2023.
- [9] David Farache, Juan Carlos Verduzco, and Alejandro Strachan. "Active learning for accelerated discovery of high-temperature MPCA". In: *MaRDA Annual Meeting*. 2022.
- [10] Ernesto Marinero, Juan Carlos Verduzco, Alejandro Strachan, Sebastian Calderon, Jeffrey Youngblood, and Alexander Wei. "Chemical manipulation of polymer structure of composite polymer electrolytes". In: *APS March Meeting Abstracts*. Vol. 2022. 2022, K10–006.
- [11] Juan Carlos Verduzco, Zachary McClure, David Farache, Saaketh Desai, and Alejandro Strachan. "Exploring the Compositional Space of High Entropy Alloys via Sequential Learning". In: *TMS2021 Technical Program*. 2021.
- [12] Juan Carlos Verduzco, Andres Villa, Alejandro Strachan, Jeffrey Youngblood, and Ernesto Marinero. "Optimizing Ionic Transport in Polymer (PEO)-Garnet (LLZO) Composite Solid-State Electrolytes". In: *APS March Meeting Abstracts*. Vol. 2021. 2021, E61–013.
- [13] Ernesto Marinero, Andres Villa, and Juan Carlos Verduzco. "High Ionic Conductivity Composite Solid-State Electrolytes". In: *Bulletin of the American Physical Society*, 65. Vol. 2020. 2020.
- [14] Juan Carlos Verduzco, Saaketh Desai, Alejandro Strachan, and Tanya Faltens. "Machine Learning for Materials Science: Open, Online Tools in NanoHUB". In: *TMS2020 Technical Program*. 2020.

HONORS AND AWARDS

Bravo Award <i>Purdue University</i>	2024
Outstanding Graduate Service Award <i>Purdue University</i>	2023
Purdue Boiler Changemaker Award <i>Purdue University</i>	2020
Doctoral Fellowship <i>Mexican National Council of Humanities, Sciences, and Technologies</i>	2018 - 2022
National Award for Excellent Performance <i>Mexican National Center for Higher Education Evaluation</i>	2017
"Santander Iberoamerica" National Scholarship <i>Santander Bank</i>	2016
National Scholarship for Academic Achievement <i>Monterrey Institute of Technology and Higher Education (ITESM)</i>	2013

TEACHING EXPERIENCE

Graduate Mentor - Undergraduate Research Programs

Fall 2018 - Summer 2023

Purdue University

- Mentored several Purdue undergraduate students on research projects combining material science, data engineering, and experimental synthesis/characterization.
- Supervised projects resulting in academic publications and presentations.

Graduate Mentor - GradTrack Program

2022

Purdue University

- Served as a mentor for the GradTrack 2022 program to support underrepresented minority students in their application process for graduate school.

Graduate Teaching Assistant - Materials Properties Laboratory (MSE 235)

Spring 2020

Purdue University

- Led laboratory sessions for a materials characterization course with 50 undergraduate students, providing instruction on various experimental techniques and data analysis methods.
- Developed and produced a video-series on materials characterization to enhance the learning experience for remote students, covering key concepts and experimental procedures in a clear and concise manner.

RELEVANT SKILLS AND LANGUAGES

Programming Languages

- Proficient: Python, LaTeX
- Familiar: C++, SQL, MATLAB
- Tools: SLURM, Git/SVN

Materials Characterization

- Techniques: XRD, DSC, EIS, SEM, Raman Spectroscopy
- Computational Packages: VASP, LAMMPS

Machine Learning and Data Science

- Frameworks: TensorFlow/Keras, PyTorch
- Libraries: Pandas, Scikit-learn, Plotly

Software

- Design: SolidWorks, CATIA
- Others: LabVIEW, Microsoft Office Suite

Languages

- English (Fluent)
- Spanish (Fluent)
- German (Beginner)

PROFESSIONAL ACTIVITIES

Workshop Organization

- **LLM Hackathon for Applications in Materials Science & Chemistry (2025)**, Argonne National Laboratories — In-person Site Organizer - West Lafayette.
- **Artificial Intelligence and Machine Learning for Materials: State of the Art and Outlook (2025)**, Purdue University, Los Alamos National Laboratories — Panelist.
- **Purdue Workshop on Digital Twins Innovation (2025)**, Purdue University — Panelist and co-organizer.
- **nanoHUB's "AI 4 Science Challenge" competition (2024)**, Purdue University — Organizer.
- **FAIR Workflows in Materials Science Workshop (2024)**, Purdue University — Technical support.
- **MaRDA Annual Meeting (2022, 2023)**, Virtual — Poster session coordinator.
- **nanoHUB Hands-on Data Science and Machine Learning Training Series (Online, 2019–present)** Purdue University — Co-organizer with Rushik Desai, Saaketh Desai and Arun Mannodi-Kanakithodi. <https://nanohub.org/groups/ml/handsontaining>

Graduate Student Organization Activities

- **Materials Science and Engineering Graduate Student Association**, Purdue University — Secretary (2019), Vice President (2020), Peer Mentoring Chair (2021).
- **Mexican and Mexican-American Association**, Purdue University — Cofounder / Treasurer (2019), President (2020–2021), Social Chair (2022), Vice President (2023).
- **Latino Graduate Student Organization**, Purdue University — President (2018–2022).

STUDENT MENTORING

Name	Supervisor	Student Status	Project Topic	Year
Lakshay Goel	Dr. Alejandro Strachan	Masters	Development of Digital Twins for Birck Nanotechnology Center	2025
Mohnish Harwani	Dr. Alejandro Strachan	Undergraduate	Applications of FAIR data for the optimization of active learning frameworks	2025
Thomas Deucher	Dr. Alejandro Strachan	Undergraduate	Impact of multi-group FAIR data on ML models using nanoHUB cyberinfrastructure	2025
Travis Jakl	Dr. Alejandro Strachan	Undergraduate (Iowa State)	Active learning techniques for the optimization of Electronic Design Automation software workflows	2025
Javier Castillo	Dr. Ernesto Marinero	Undergraduate	Development of Composite Polymer Electrolytes based on Lithium-doped High-entropy Oxides	2023
Yaisa Ramirez	Dr. Ernesto Marinero	Undergraduate	Development of Composite Polymer Electrolytes based on Lithium-doped High-entropy Oxides	2023
Kaiqi Zhang	Dr. Ernesto Marinero	Undergraduate	Synthesis of colossal dielectric constant high-entropy ceramics	2023
David Emdin	Dr. Tanya Faltens	Undergraduate (Community College of Philadelphia)	Introduction to MD Simulations (Mean Squared Displacement)	2022
David E. Farache	Dr. Alejandro Strachan	Undergraduate	Active learning strategies for the discovery of multiple principal component alloys for high temperature applications	2022
Gavin Bidna	Dr. Ernesto Marinero	Undergraduate	Synthesis and Processing of highly ionically conductive High-Entropy Oxides	2022
Gabrielle Gushurst	Dr. Ernesto Marinero	Masters	Synthesis of High-entropy Oxides for Water-Splitting	2022
George M. Nishibuchi	Dr. Alejandro Strachan	Undergraduate	Li-ion diffusion MD studies on doped Lithium Lanthanum Zirconium Oxide	2021
Raina A Shreiner Barger	Dr. Ernesto Marinero	Undergraduate	Effects of plasticizer Succinonitrile in Composite Polymer Electrolytes	2021
John N. Vergados	Dr. Ernesto Marinero	Undergraduate	Synthesis of polymer electrolytes for solid-state batteries	2021
Yana Johnson	Dr. Alejandro Strachan	Undergraduate (Dillard)	Evaluation of student experience on Machine Learning modules in nanoHUB	2021
David Starr	Dr. Ernesto Marinero	Undergraduate	Development of Composite Polymer Electrolytes based on doped Lithium Lanthanum Zirconium Oxide	2020
Julia K. Peck	Dr. Ernesto Marinero	Undergraduate	Development of Composite Polymer Electrolytes based on doped Lithium Lanthanum Zirconium Oxide	2020
Noah Hallberg	Dr. Alejandro Strachan	High School	Design and deployment of a Wordpress research website	2020
Haydn Schroader	Dr. Alejandro Strachan	Undergraduate	Development of introductory Machine Learning modules for Materials Science in nanoHUB	2019